

Nithin Ram Kalava

Software Engineer

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SUMMARY

Software engineer at SecureMachines building an enterprise Cloud HSM: the cloud services and infrastructure that let organizations use hardware security modules remotely, compatible with existing KMS, SignServer and PKCS#11 tooling. Work spans cloud and on-prem, from a JCE provider and C shared libraries up to GCP infrastructure. Author of pqc, a post-quantum cryptography library used by other developers.

EXPERIENCE

SecureMachines

Software Engineer

Bengaluru, India

Nov 2025 - Present

- Building SecureMachines' enterprise Cloud HSM: the cloud services and infrastructure that give organizations remote access to hardware HSMs while staying compatible with existing KMS, SignServer and PKCS#11 tooling, across both cloud and on-prem. Also built a lightweight PKCS#11-over-REST API as an exploratory enabler for trying the post-quantum HSM.
- Primary engineer on national identity integrations: integrating the company's HSMs into a national-scale biometric identity system, and into an open-source identity platform that governments deploy.
- Designed and implemented components of Augha Entropy-as-a-Service (EaaS), a customer-facing cryptographic platform on Google Cloud, using Compute Engine, Cloud Run, Cloud KMS, Firestore, Memorystore, API Gateway, load balancers and HA VPN.
- Brought hardware HSM support to Linux by developing C shared libraries (.so) and integrating PKCS#11 for cryptographic operations; built a Java Cryptography Extension (JCE) provider so existing Java applications use the HSMs with no code changes.
- Improved cryptographic performance from about 1.8 seconds for 256-byte workloads to about 11 ms for 10,000-byte workloads through protocol and implementation optimizations.
- Automated FPGA board programming and validation using Tcl scripts, reducing manual flashing effort and improving development efficiency.

SELECTED PROJECTS

pqc | *Post-quantum cryptography for JavaScript*

npmjs.com/package/pqc

- Pure-JavaScript implementation of the NIST post-quantum standards (ML-KEM, ML-DSA, SLH-DSA; FIPS 203, 204, 205), with no native dependencies.
- Over 3,400 npm downloads. More than 2,300 ML-KEM key generations per second in JavaScript. Accompanied by a research paper.

PQC-Vizz | *Next.js, TypeScript, SVG*

pqc.nithinram.com

- Interactive, in-browser visualizer for post-quantum algorithms that runs the real pqc library live.

PC Building Assistant | *PostgreSQL, PL/pgSQL, ML*

pc.nithinram.com

- LLM and ML component recommender with a PostgreSQL compatibility engine (PL/pgSQL), gradient-boosting ranking models, and an ETL pipeline.

CareerPath Navigator | *Next.js*

careerpath-nav.nithinram.com

- Career-exploration platform for rural students, optimized for low-bandwidth devices and networks. Accessibility-focused.

SKILLS

Languages: Java, C, Python, TypeScript, SQL, Bash, Tcl

Cryptography & Security: HSMs, PKCS#11, JCE, KMS, SignServer, post-quantum (ML-KEM, ML-DSA, SLH-DSA)

Cloud & Infrastructure: Google Cloud (Compute Engine, Cloud Run, Cloud KMS, Firestore, Memorystore, API Gateway, HA VPN), AWS, Linux, Docker, CI/CD

Web & Data: Next.js, React, Node.js, PostgreSQL, REST APIs, Pandas

EDUCATION

Vasireddy Venkatadri Institute of Technology

B.Tech, Computer Science

Guntur, Andhra Pradesh, India

2021 - Jun 2025

CERTIFICATIONS

AWS Academy and Google Cloud training; a selection of machine-learning and data-science certificates (DeepLearning.AI, IBM).